Possible outcomes and equivalent formulations

Sapienza Università di Roma

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The two reference forms

Introduction: The two reference forms

- Throughout this chapter we refer to the two forms of an LP problem introduced in the previous chapter: the **canonical** and the **standard maximization form**
- We recall them, together with the three LP problems used as guiding examples
- They are the reference point of the whole chapter: we will see which outcomes an LP problem can have, and how every LP problem can be brought into one of these two forms

The data of an LP problem

- Both forms are written with the same data:
  - an **objective-function vector** $\boldsymbol{c} \in \mathbb{R}^n$;
  - a **constraint matrix** $\boldsymbol{A} \in \mathbb{R}^{m \times n}$;
  - a **right-hand-side vector** $\boldsymbol{b} \in \mathbb{R}^m$.
- The two forms differ only in the type of the m linear constraints: "$\leq$" in the canonical form, "$=$" in the standard form

Canonical maximization form

An LP problem in **canonical maximization form** with n non-negative variables and m less-than-or-equal-to linear constraints:

$$\max \quad \sum_{j=1}^{n} c_j \, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} \, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\}.$$

In matrix notation:

$$\max \quad \boldsymbol{c}' \boldsymbol{x}$$

$$\text{subject to} \quad \boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b},$$

$$\boldsymbol{x} \geq 0.$$

Standard maximization form

An LP problem in **standard maximization form** with n non-negative variables and m equality linear constraints:

$$\max \quad \sum_{j=1}^{n} c_j \, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} \, x_j = b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\}.$$

In matrix notation:

$$\max \quad \boldsymbol{c}' \boldsymbol{x}$$

$$\text{subject to} \quad \boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b},$$

$$\boldsymbol{x} \geq 0.$$

We also refer to the three LP problems
used as **guiding examples** in the previous chapter,

which we recall in the next three slides.

Example: first guiding LP problem

The first guiding example is the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints:

$$\begin{array}{rrcrcrcr} \max & 5 & x_1 & + & 2 & x_2 & & \\ \text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\ & & x_1 & + & 2 & x_2 & \leq & 12 \\ & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

An optimal solution is $\tilde{\boldsymbol{x}}(\, LP\,) = \begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix}$ and the optimal value is $z(\, LP\,) = 32$.

Example: second guiding LP problem

The second guiding example is the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 2$ less-than-or-equal-to linear constraints:

$$\begin{array}{rrrrrrrr} \max & -2 & x_1 & - & 4 & x_2 & & \\ \text{subject to} & - & x_1 & - & & x_2 & \leq & -2 \\ & & x_1 & - & & x_2 & \leq & 0 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

An optimal solution is $\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and the optimal value is $z(\,LP\,) = -6$.

Example: third guiding LP problem

The third guiding example is the following LP problem in standard maximization form with $n = 3$ non-negative variables and $m = 1$ equality linear constraint:

$$\begin{array}{rrrrrrrrrr} \max & 2 & x_1 & + & 2 & x_2 & + & x_3 & & \\ \text{subject to} & 2 & x_1 & + & 3 & x_2 & + & x_3 & = & 4 \\ & & x_1, & & & x_2, & & x_3 & \geq & 0 \end{array}$$

Its optimal value is $z(\,LP\,) = 4$, attained by every point of the segment joining $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}$.

Summary

Canonical maximization form:

- $\max \boldsymbol{c}'\boldsymbol{x}$ subject to $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{x} \geq 0$: n non-negative variables, m "$\leq$" constraints

Standard maximization form:

- $\max \boldsymbol{c}'\boldsymbol{x}$ subject to $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq 0$: n non-negative variables, m "$=$" constraints

Three guiding examples:

- two in canonical form (optimal values 32 and -6), one in standard form (optimal value 4, attained by a whole segment)

Infeasible and unbounded LP problems

Introduction: Infeasible and unbounded LP problems

- The three guiding LP problems admit an optimal solution, but this is not always the case
- We define **infeasible** and **unbounded** LP problems, and show that every LP problem has exactly one of three possible outcomes
- We see one example for each case, with its picture in $\mathbb{R}^2$
- Finally, we show that an LP problem with an optimal solution has either exactly one or infinitely many optimal solutions

Not always an optimal solution

- The LP problems of the three guiding examples admit an optimal solution
- This is not always the case: an LP problem may have
  - no feasible solution at all, or
  - feasible solutions of arbitrarily large objective value

Infeasible LP problem

Definition 1: infeasible LP problem

An LP problem is called ***infeasible*** *if its feasible region is empty:*

$$F = \emptyset.$$

Unbounded LP problem

Definition 2: unbounded LP problem

An LP problem in ***maximization form*** *is called* ***unbounded*** *if, for every value* $M \in \mathbb{R}$*, there exists a feasible solution* $\bar{\boldsymbol{x}}(\, LP\,) \in F$ *such that*

$$f(\bar{\boldsymbol{x}}(\, LP\,)) > M.$$

An LP problem in ***minimization form*** *is called* ***unbounded*** *if, for every value* $M \in \mathbb{R}$*, there exists a feasible solution* $\bar{\boldsymbol{x}}(\, LP\,) \in F$ *such that* $f(\bar{\boldsymbol{x}}(\, LP\,)) < M.$

The three possible outcomes

An LP problem satisfies **exactly one** of the following three conditions:

1. it is **infeasible**;
2. it is **unbounded**;
3. it admits an **optimal solution** $\tilde{\boldsymbol{x}}(\, LP\,)$, and the optimal value $z(\, LP\,)$ is a finite number.

The optimal value in the first two cases

- An **infeasible** LP problem has no feasible solution, so it has no optimal solution and the optimal value $z(\,LP\,)$ is not defined
- An **unbounded** LP problem has feasible solutions, but none of them is optimal, and the optimal value $z(\,LP\,)$ is not defined either
- In the two cases, with a slight abuse of notation, we write:
  - $z(\,LP\,) = -\infty$ for an infeasible LP problem in maximization form;
  - $z(\,LP\,) = +\infty$ for an unbounded LP problem in maximization form.

Let us now see one example of each case:

an **infeasible** LP problem,
an **unbounded** LP problem,
and an unbounded feasible region with a **finite** optimal value.

Example: infeasible LP problem (1/2)

Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 2$ less-than-or-equal-to linear constraints:

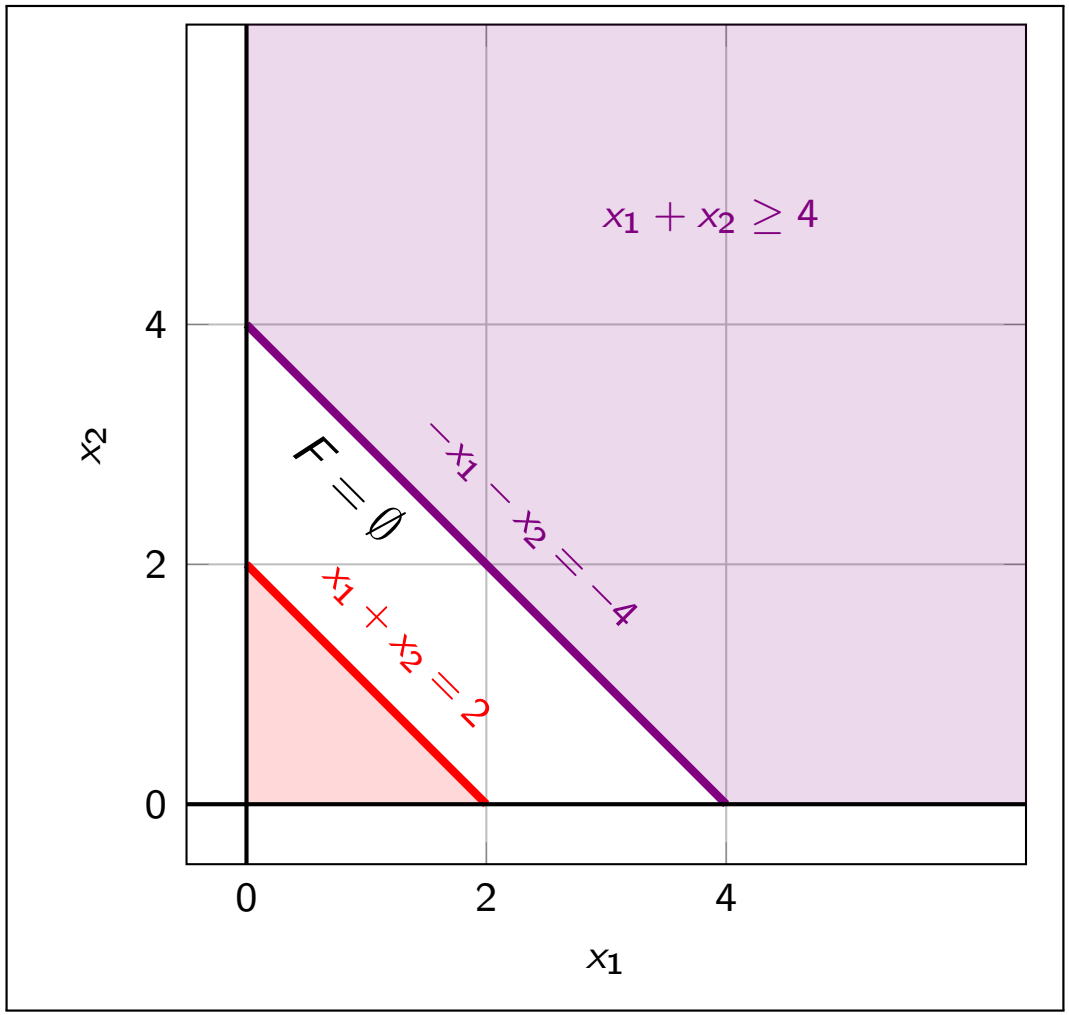
$$\begin{array}{rrcrcr} \max & x_1 & + & x_2 & & \\ \text{subject to} & x_1 & + & x_2 & \leq & 2 \\ - & x_1 & - & x_2 & \leq & -4 \\ & x_1, & & x_2 & \geq & 0 \end{array}$$

- The first constraint imposes $x_1 + x_2 \leq 2$ and the second one imposes $x_1 + x_2 \geq 4$
- No point of $\mathbb{R}^2$ satisfies both of them, and then we have:

$$F = \emptyset.$$

- Accordingly, this LP problem is **infeasible**

Example: infeasible LP problem (2/2)



- In red the half-plane $x_1 + x_2 \leq 2$ (with the non-negativity constraints)
- In violet the half-plane $x_1 + x_2 \geq 4$
- Their intersection is **empty**

Example: unbounded LP problem (1/3)

Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 2$ less-than-or-equal-to linear constraints:

$$
\begin{array}{rrrrrrr}
\max & & x_1 & + & x_2 & & \\
\text{subject to} & - & x_1 & + & x_2 & \leq & 1 \\
 & & x_1 & - & x_2 & \leq & 1 \\
 & & x_1, & & x_2 & \geq & 0
\end{array}
$$

Example: unbounded LP problem (2/3)

- For every value $t \geq 0$, the point $\bar{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} t \\ t \end{pmatrix}$ is a feasible solution, since

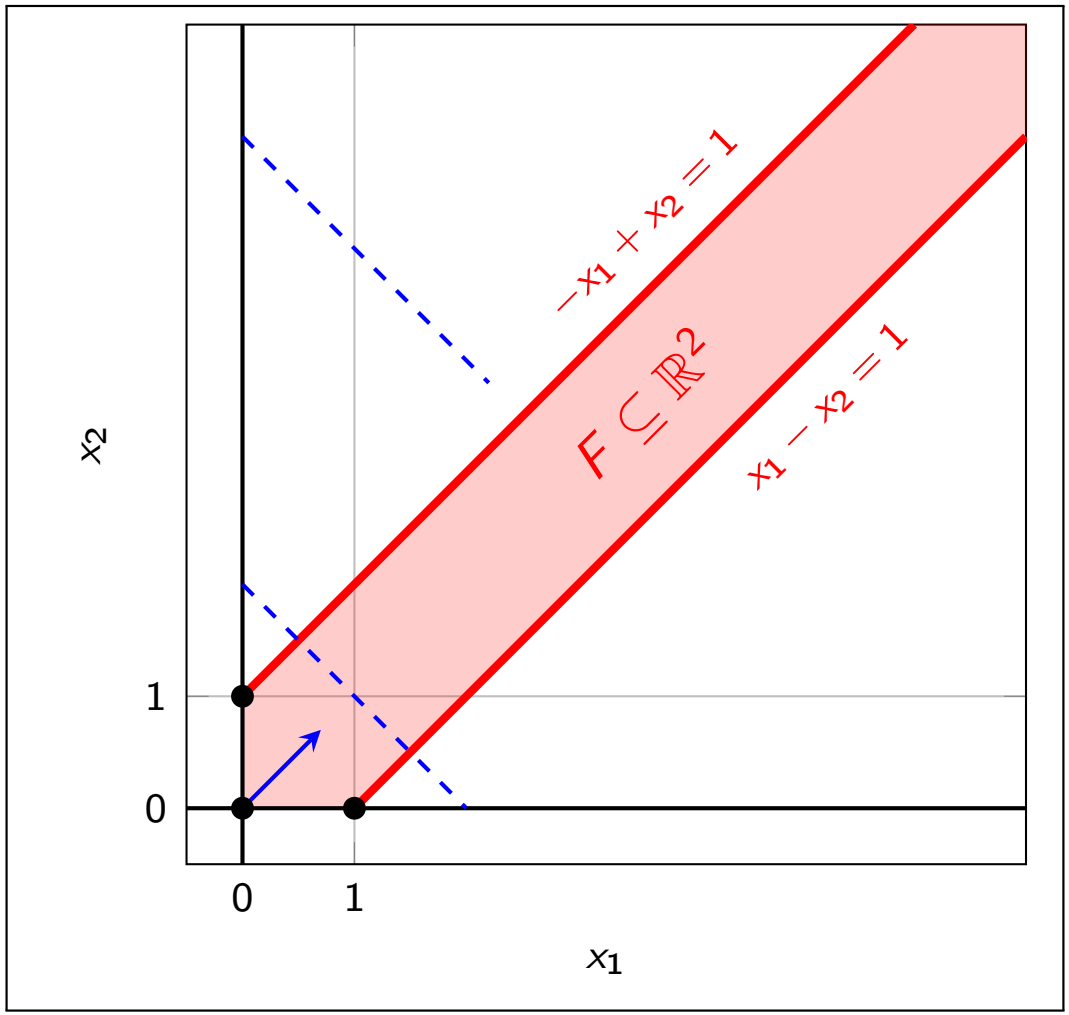
$$-t + t = 0 \leq 1, \qquad t - t = 0 \leq 1, \qquad t \geq 0 \quad \text{and} \quad t \geq 0,$$

- and its objective value is

$$LB(\bar{\boldsymbol{x}}) = t + t = 2\,t.$$

- Accordingly, for every value $M \in \mathbb{R}$, the feasible solution obtained with $t > \frac{M}{2}$ has an objective value larger than M, and then this LP problem is **unbounded**

Example: unbounded LP problem (3/3)



- In dashed blue the level sets $x_1 + x_2 = 2$ and $x_1 + x_2 = 6$ of the objective function
- The gradient points along the unbounded feasible region

Unbounded region, not unbounded problem

An LP problem whose feasible region F is unbounded
is **not necessarily** an unbounded LP problem:

the objective function may well attain
a **finite optimal value** on F.

- The next example shows such a case

Example: unbounded feasible region and multiple optimal solutions (1/3)

Consider the following LP problem in canonical maximization form with $n = 2$ non-negative variables and $m = 2$ less-than-or-equal-to linear constraints:

$$\begin{array}{rrrrrrr} \max & - & x_1 & - & x_2 & & \\ \text{subject to} & - & x_1 & - & x_2 & \leq & -2 \\ & & x_1 & - & x_2 & \leq & 2 \\ & & x_1, & & x_2 & \geq & 0 \end{array}$$

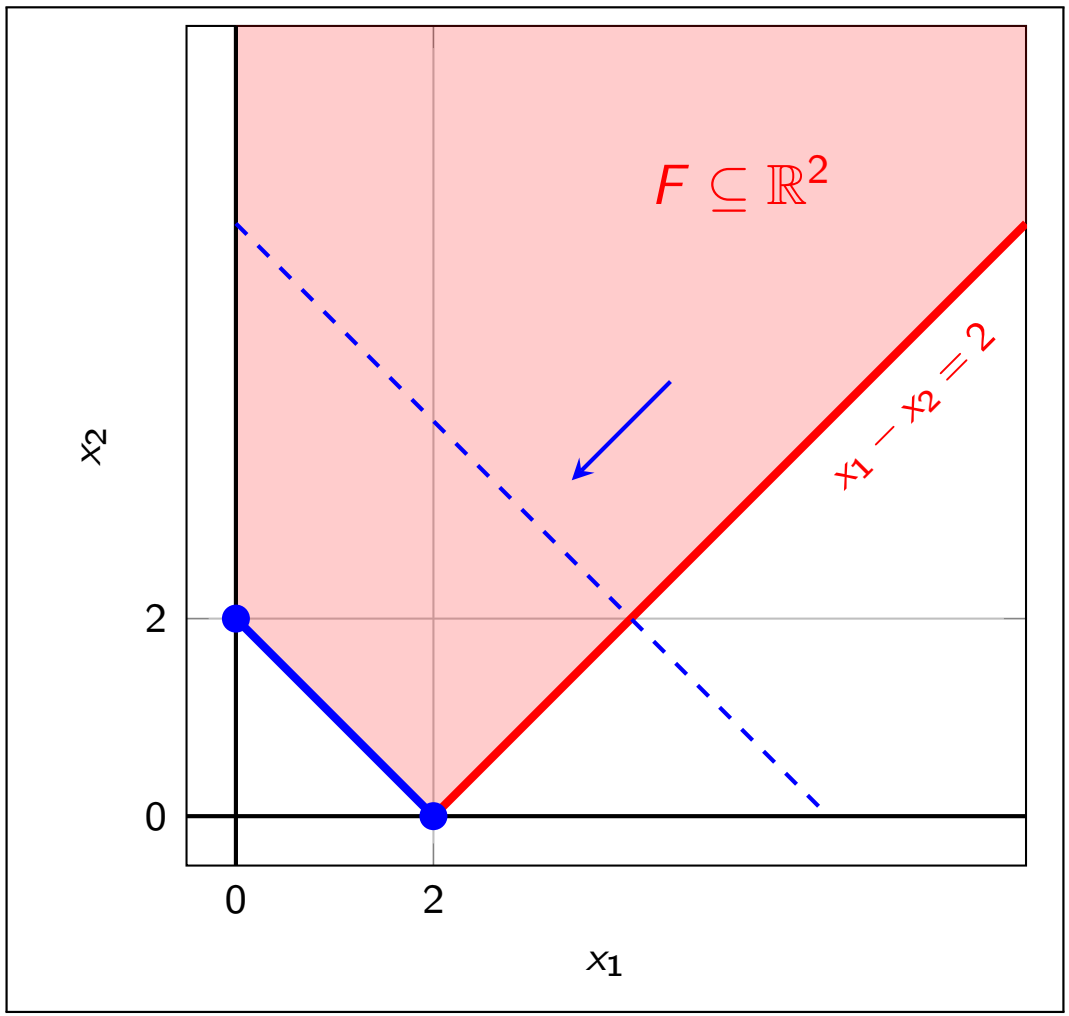
Example: unbounded feasible region and multiple optimal solutions (2/3)

- The feasible region F is unbounded
- Nevertheless, the first constraint imposes $x_1 + x_2 \geq 2$ for every feasible solution, and then

$$-x_1 - x_2 \leq -2, \qquad \forall \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in F.$$

- The feasible solutions $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ attain this value
- Then they are two **optimal solutions** and the optimal value is $z(\,LP\,) = -2$

Example: unbounded feasible region and multiple optimal solutions (3/3)



- In dashed blue the level set $-x_1 - x_2 = -6$
- In solid blue the level set $-x_1 - x_2 = -2$, which contains the segment of optimal solutions joining the two blue vertices

In the last example the optimal solutions are infinitely many: the whole segment joining the two blue vertices.

This is a general fact.

Multiple optimal solutions

Observation 1: multiple optimal solutions

If $\tilde{\boldsymbol{x}}^{(1)}(\,LP\,)$ and $\tilde{\boldsymbol{x}}^{(2)}(\,LP\,)$ are two optimal solutions of an LP problem, then every point

$$\lambda\,\tilde{\boldsymbol{x}}^{(1)}(\,LP\,)+(1-\lambda)\,\tilde{\boldsymbol{x}}^{(2)}(\,LP\,),\qquad \lambda\in[0,1],$$

is an optimal solution as well. Accordingly, an LP problem that admits an optimal solution admits either exactly one optimal solution or infinitely many optimal solutions.

Proof (1/2)

- Let $\boldsymbol{x}^{\lambda} = \lambda\, \tilde{\boldsymbol{x}}^{(1)}(\, LP\,) + (1-\lambda)\, \tilde{\boldsymbol{x}}^{(2)}(\, LP\,)$ with $\lambda \in [0,1]$
- First, $\boldsymbol{x}^{\lambda} \in F$. Indeed, the feasible region F is a polyhedron and, for each of its inequalities $\boldsymbol{a}\,\boldsymbol{x} \leq b$, since $\lambda \geq 0$, $1-\lambda \geq 0$ and both solutions satisfy it, we have:

$$\begin{aligned} \boldsymbol{a}\,\boldsymbol{x}^{\lambda} &= \lambda\, \boldsymbol{a}\,\tilde{\boldsymbol{x}}^{(1)}(\, LP\,) + (1-\lambda)\, \boldsymbol{a}\,\tilde{\boldsymbol{x}}^{(2)}(\, LP\,) \\ &\leq \lambda\, b + (1-\lambda)\, b = b; \end{aligned}$$

- The same computation, with $=$ in place of $\leq$, holds for each of its equalities

Proof (2/2)

- Moreover, since the objective function f is linear, we have:

$$f(\boldsymbol{x}^{\lambda}) = \lambda\, f(\tilde{\boldsymbol{x}}^{(1)}(\,LP\,)) + (1-\lambda)\, f(\tilde{\boldsymbol{x}}^{(2)}(\,LP\,))$$

$$= \lambda\, z(\,LP\,) + (1-\lambda)\, z(\,LP\,) = z(\,LP\,),$$

  where the second equality holds because both solutions are optimal

- Accordingly, $\boldsymbol{x}^{\lambda}$ is a feasible solution whose objective value is the optimal value, and then it is an optimal solution

□

Summary

Three possible outcomes:

- an LP problem is **infeasible** ($F = \emptyset$), **unbounded** (objective value arbitrarily large), or it admits an **optimal solution** with a finite optimal value
- we write $z(\,LP\,) = -\infty$ (infeasible) and $z(\,LP\,) = +\infty$ (unbounded) for a maximization problem

Unbounded region $\neq$ unbounded problem:

- an unbounded feasible region may still give a finite optimal value

Multiple optimal solutions:

- every convex combination of two optimal solutions is optimal: exactly one or infinitely many

Equivalent reformulations

Introduction: Equivalent reformulations

- An LP problem met in practice is rarely written in canonical or standard maximization form
- We show that every **generic LP problem** can be rewritten in both forms, with simple transformations of the objective function, of the constraints and of the variables
- This matters because the methods of the next chapters, starting from the simplex algorithm, work on one of these two forms
- At the end, we also see that rational input data can be replaced by integer input data

Generic LP problems

- A **generic LP problem** may be written in a form which is neither the canonical maximization form nor the standard maximization form:
  - its objective function may have to be **minimized**;
  - its constraints may be of the three types "$\leq$", "$\geq$" and "$=$";
  - its variables may be **non-positive** or **free**.
- In this section we show that every generic LP problem can be rewritten in canonical maximization form and in standard maximization form

Equivalent LP problems

Definition 3: equivalent LP problems

Two LP problems are called ***equivalent*** *if, from an optimal solution of one of them, an optimal solution of the other one can be directly obtained, and the two problems are both infeasible, both unbounded, or both admit an optimal solution.*

We start from the objective function:

how to turn a **minimization** into a **maximization**.

Minimization and maximization form

- For every linear function f and every polyhedron F, we have:

$$\min\{\, f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}$$

$$= -\max\{\, -f(\boldsymbol{x}) \text{ subject to } \boldsymbol{x} \in F \,\}.$$

- Accordingly, an LP problem in **minimization form** with objective-function vector $\boldsymbol{c}$ is equivalent to the LP problem in **maximization form** with the same feasible region and objective-function vector $-\boldsymbol{c}$:

$$\underbrace{\sum_{j=1}^{n} c_j\, x_j}_{\boldsymbol{c}'\boldsymbol{x}} \quad \longrightarrow \quad \underbrace{\sum_{j=1}^{n} (-c_j)\, x_j}_{-\boldsymbol{c}'\boldsymbol{x}} .$$

Same optimal solutions, opposite optimal values

The two equivalent LP problems have
the **same optimal solutions**,
while their optimal values are opposite numbers:

$$\tilde{\boldsymbol{x}}(\, LP_{\mathsf{min}}\,) = \tilde{\boldsymbol{x}}(\, LP_{\mathsf{max}}\,) \qquad \text{and} \qquad z(\, LP_{\mathsf{min}}\,) = -\, z(\, LP_{\mathsf{max}}\,).$$

Example: from the minimization to the maximization form (1/2)

Consider the following LP problem in minimization form with $n = 2$ non-negative variables:

$$\begin{array}{rrrrrrrr} \min & 2 & x_1 & + & 4 & x_2 & & \\ \text{subject to} & - & x_1 & - & & x_2 & \leq & -2 \\ & & x_1 & - & & x_2 & \leq & 0 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

- We change the sign of the objective-function vector

Example: from the minimization to the maximization form (2/2)

By changing the sign of the objective-function vector, we obtain the equivalent LP problem in canonical maximization form of the second guiding example:

$$\begin{array}{rrrrrrrr} \max & -2 & x_1 & - & 4 & x_2 & & \\ \text{subject to} & - & x_1 & - & & x_2 & \leq & -2 \\ & & x_1 & - & & x_2 & \leq & 0 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

- The optimal solution of both problems is $\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- The optimal value of the minimization form is 6 and the optimal value of the maximization form is -6

Now the constraints: how to turn
"$\geq$" and "$=$" constraints into "$\leq$" constraints.

Greater-than-or-equal-to constraints

- A **greater-than-or-equal-to** linear constraint is equivalent to a less-than-or-equal-to linear constraint, since, by multiplying both sides by -1, we have:

$$\sum_{j=1}^{n} a_{ij}\, x_j \geq b_i \qquad \Longleftrightarrow \qquad \sum_{j=1}^{n} (-a_{ij})\, x_j \leq -b_i .$$

Equality constraints

- An **equality** linear constraint is equivalent to two less-than-or-equal-to linear constraints, since:

$$\sum_{j=1}^{n} a_{ij}\, x_j = b_i \quad \Longleftrightarrow \quad \underbrace{\sum_{j=1}^{n} a_{ij}\, x_j \leq b_i}_{\text{first constraint}} \quad \text{and} \quad \underbrace{\sum_{j=1}^{n} (-a_{ij})\, x_j \leq -b_i}_{\text{second constraint}} .$$

From the standard to the canonical maximization form

- An LP problem in standard maximization form with n non-negative variables and m equality linear constraints is equivalent to an LP problem in canonical maximization form with the same n non-negative variables and $2\,m$ less-than-or-equal-to linear constraints
- Each equality constraint is replaced by its two less-than-or-equal-to constraints (next slide)
- The two LP problems have the **same feasible region**, and then the same optimal solutions and the same optimal value

From the standard to the canonical maximization form: the LP problem

The equivalent LP problem in canonical maximization form with n non-negative variables and $2\,m$ less-than-or-equal-to linear constraints:

$$\max \quad \sum_{j=1}^{n} c_j\, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij}\, x_j \leq b_i, \qquad \forall i \in \{1, 2, \ldots, m\},$$

$$\sum_{j=1}^{n} (-a_{ij})\, x_j \leq -b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, n\}.$$

Example: from the standard to the canonical maximization form (1/2)

Consider the LP problem in standard maximization form of the third guiding example, with $n = 3$ non-negative variables and $m = 1$ equality linear constraint:

$$\begin{array}{rrrrrrrrrr} \max & 2 & x_1 & + & 2 & x_2 & + & x_3 & & \\ \text{subject to} & 2 & x_1 & + & 3 & x_2 & + & x_3 & = & 4 \\ & & x_1, & & & x_2, & & x_3 & \geq & 0 \end{array}$$

- We replace the equality linear constraint with two less-than-or-equal-to linear constraints

Example: from the standard to the canonical maximization form (2/2)

By replacing the equality linear constraint with two less-than-or-equal-to linear constraints, we obtain the equivalent LP problem in canonical maximization form with $n = 3$ non-negative variables and $2\,m = 2$ less-than-or-equal-to linear constraints:

$$
\begin{array}{rrcrcrcrr}
\max & 2 & x_1 & + & 2 & x_2 & + & x_3 & \\
\text{subject to} & 2 & x_1 & + & 3 & x_2 & + & x_3 & \leq & 4 \\
 & -2 & x_1 & - & 3 & x_2 & - & x_3 & \leq & -4 \\
 & & x_1, & & & x_2, & & x_3 & \geq & 0
\end{array}
$$

The opposite direction: how to turn
"$\leq$" and "$\geq$" constraints into "$=$" constraints,
by adding new variables.

Slack and surplus variables

- A **less-than-or-equal-to** linear constraint is equivalent to an equality linear constraint and a non-negativity constraint, obtained by adding one non-negative variable x_{n+i}:

$$\sum_{j=1}^{n} a_{ij}\, x_j \leq b_i \quad \Longleftrightarrow \quad \sum_{j=1}^{n} a_{ij}\, x_j + x_{n+i} = b_i \quad \text{and} \quad x_{n+i} \geq 0.$$

- A **greater-than-or-equal-to** linear constraint is equivalent to an equality linear constraint and a non-negativity constraint, obtained by subtracting one non-negative variable x_{n+i}:

$$\sum_{j=1}^{n} a_{ij}\, x_j \geq b_i \quad \Longleftrightarrow \quad \sum_{j=1}^{n} a_{ij}\, x_j - x_{n+i} = b_i \quad \text{and} \quad x_{n+i} \geq 0.$$

Slack and surplus variables

Definition 4: slack and surplus variables

*The additional non-negative variable x_{n+i} is called the **slack variable** of a less-than-or-equal-to linear constraint and the **surplus variable** of a greater-than-or-equal-to linear constraint. Its value measures, for a given feasible solution, the difference between the two sides of the constraint.*

From the canonical to the standard maximization form

An LP problem in canonical maximization form with n non-negative variables and m less-than-or-equal-to linear constraints is equivalent to an LP problem in standard maximization form with $n + m$ non-negative variables and the same m linear constraints, now equalities:

$$\max \quad \sum_{j=1}^{n} c_j \, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} \, x_j + x_{n+i} = b_i, \qquad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n + m\}.$$

The standard form in matrix notation

- In matrix notation, the m slack variables form a column vector $\boldsymbol{x}_S \in \mathbb{R}^m_{\geq 0}$
- The constraint matrix of the standard maximization form is obtained by appending to $\boldsymbol{A}$ the **identity matrix** $\boldsymbol{I}$ of size m:

$$\max \quad \boldsymbol{c}' \boldsymbol{x}$$

$$\text{subject to} \quad \underbrace{\boldsymbol{A}\,\boldsymbol{x} + \boldsymbol{I}\,\boldsymbol{x}_S}_{(\boldsymbol{A} \quad \boldsymbol{I})\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{x}_S\end{pmatrix}} = \boldsymbol{b},$$

$$\boldsymbol{x},\ \boldsymbol{x}_S \geq 0.$$

The slack vector and the identity matrix

In the matrix form of the previous slide we have:

$$\boldsymbol{x}_S = \begin{pmatrix} x_{n+1} \\ x_{n+2} \\ \vdots \\ x_{n+m} \end{pmatrix} \qquad \boldsymbol{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

Example: from the canonical to the standard maximization form (1/2)

By adding $m = 3$ slack variables, the LP problem of the first guiding example becomes the following equivalent LP problem in standard maximization form with $n + m = 5$ non-negative variables:

$$
\begin{array}{rrrrrrrrrrrrrrr}
\max & 5 & x_1 & + & 2 & x_2 & & & & & & & & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & + & x_3 & & & & & & = & 12 \\
 & & x_1 & + & 2 & x_2 & & & + & x_4 & & & & = & 12 \\
 & 3 & x_1 & - & 2 & x_2 & & & & & + & x_5 & = & & 8 \\
 & & x_1, & & & x_2, & & x_3, & & x_4, & & x_5 & \geq & & 0
\end{array}
$$

Example: from the canonical to the standard maximization form (2/2)

- The optimal solution of the first guiding example is

$$\tilde{\boldsymbol{x}}(\,LP\,) = \begin{pmatrix} 5 \\ \frac{7}{2} \end{pmatrix}$$

- It satisfies the first constraint strictly and the other two with equality

- Then the values of the three slack variables are

$$\tilde{x}_3 = \frac{13}{2}, \qquad \tilde{x}_4 = 0 \qquad \text{and} \qquad \tilde{x}_5 = 0.$$

Finally the variables: how to replace
non-positive and **free** variables
with non-negative ones.

Non-positive and free variables

- A **non-positive** variable $x_j \leq 0$ is equivalent to one non-negative variable $x_j^+ \geq 0$, by setting:

$$x_j = -x_j^+.$$

- A **free** variable $x_j \gtreqless 0$ is equivalent to two non-negative variables $x_j^+ \geq 0$ and $x_j^- \geq 0$, by setting:

$$x_j = x_j^+ - x_j^-.$$

Where the substitution is performed

- In both cases, the substitution is performed in the **objective function** and in **every linear constraint** of the LP problem
- A free variable is replaced by two non-negative variables, since every real number is the difference of two non-negative numbers

Example: equivalent LP problem with only non-negative variables (1/2)

Consider the following LP problem with one non-negative variable $x_1 \geq 0$, one non-positive variable $x_2 \leq 0$ and one free variable $x_3 \gtreqless 0$:

$$
\begin{array}{rrrrrrrrrrr}
\max & 2 & x_1 & + & 4 & x_2 & + & 2 & x_3 & & \\
\text{subject to} & - & x_1 & + & & x_2 & + & & x_3 & = & 2 \\
& 8 & x_1 & + & 2 & x_2 & & & & = & 19 \\
& & x_1 & & & & & & & \geq & 0 \\
& & & & & x_2 & & & & \leq & 0 \\
& & & & & & & & x_3 & \gtreqless & 0
\end{array}
$$

Example: equivalent LP problem with only non-negative variables (2/2)

By setting $x_2 = -x_2^+$ and $x_3 = x_3^+ - x_3^-$, we obtain the following equivalent LP problem with only non-negative variables:

$$\begin{array}{rrrrrrrrrrrrr} \max & 2 & x_1 & - & 4 & x_2^+ & + & 2 & x_3^+ & - & 2 & x_3^- & \\ \text{subject to} & - & x_1 & - & & x_2^+ & + & & x_3^+ & - & & x_3^- & = & 2 \\ & 8 & x_1 & - & 2 & x_2^+ & & & & & & & = & 19 \\ & & x_1, & & & x_2^+, & & & x_3^+, & & & x_3^- & \geq & 0 \end{array}$$

From an optimal solution of the second LP problem, an optimal solution of the first one is obtained by computing $\tilde{x}_2 = -\tilde{x}_2^+$ and $\tilde{x}_3 = \tilde{x}_3^+ - \tilde{x}_3^-$.

We now put all the transformations together in a single table.

From a generic LP problem to the two maximization forms

Every generic LP problem can be rewritten in canonical maximization form and in standard maximization form by applying the following transformations:

	canonical max form	**standard max form**
$\min \boldsymbol{c}'\boldsymbol{x}$	$-\max(-\boldsymbol{c})'\boldsymbol{x}$	$-\max(-\boldsymbol{c})'\boldsymbol{x}$
$\sum_j a_{ij}x_j \leq b_i$	unchanged	add a slack variable $x_{n+i} \geq 0$
$\sum_j a_{ij}x_j \geq b_i$	multiply by -1	subtract a surplus variable $x_{n+i} \geq 0$
$\sum_j a_{ij}x_j = b_i$	split into two "$\leq$", the second one multiplied by -1	unchanged
$x_j \geq 0$	unchanged	unchanged
$x_j \leq 0$	$x_j = -x_j^+$	$x_j = -x_j^+$
$x_j \gtreqless 0$	$x_j = x_j^+ - x_j^-$	$x_j = x_j^+ - x_j^-$

Example: from a generic LP problem to the canonical maximization form (1/3)

Consider the following generic LP problem with 3 variables:

$$
\begin{array}{rrrcrrrcrcl}
\min & 3 & x_1 & - & 2 & x_2 & + & x_3 & & & \\
\text{subject to} & & x_1 & + & & x_2 & - & x_3 & \geq & 4 \\
& 2 & x_1 & - & & x_2 & & & = & 6 \\
& & x_1 & & & & & & \geq & 0 \\
& & & & & x_2 & & & \leq & 0 \\
& & & & & & & x_3 & \lessgtr & 0
\end{array}
$$

Example: from a generic LP problem to the canonical maximization form (2/3)

We apply the transformations of the table:

1. the objective function is minimized, and then we maximize the opposite objective function;
2. the first constraint is a greater-than-or-equal-to constraint, and then we multiply it by -1;
3. the second constraint is an equality constraint, and then we split it into two less-than-or-equal-to constraints;
4. the variable x_2 is non-positive, and then we set $x_2 = -x_2^+$ with $x_2^+ \geq 0$;
5. the variable x_3 is free, and then we set $x_3 = x_3^+ - x_3^-$ with $x_3^+, x_3^- \geq 0$.

Example: from a generic LP problem to the canonical maximization form (3/3)

We obtain the following equivalent LP problem in canonical maximization form with $n = 4$ non-negative variables and $m = 3$ less-than-or-equal-to linear constraints:

$$
\begin{array}{rrcrcrcrcl}
-\max & -3 & x_1 & - & 2\,x_2^+ & - & x_3^+ & + & x_3^- & \\
\text{subject to} & - & x_1 & + & x_2^+ & + & x_3^+ & - & x_3^- & \leq -4 \\
& 2 & x_1 & + & x_2^+ & & & & & \leq 6 \\
& -2 & x_1 & - & x_2^+ & & & & & \leq -6 \\
& & x_1, & & x_2^+, & & x_3^+, & & x_3^- & \geq 0
\end{array}
$$

Finally, by adding $m = 3$ slack variables x_5, x_6, x_7, we obtain the equivalent LP problem in standard maximization form with $n + m = 7$ non-negative variables and $m = 3$ equality linear constraints.

A last equivalence concerns the input data:

rational data can always be replaced by **integer** data.

Least Common Denominator

Definition 5: Least Common Denominator

Given a vector $\boldsymbol{v} \in \mathbb{Q}^k$ *of* k *rational numbers* $v_j = \frac{p_j}{q_j}$ *with* $p_j \in \mathbb{Z}$ *and* $q_j \in \mathbb{N}$ *coprime,* $\forall j \in \{1, 2, \ldots, k\}$*, the* ***Least Common Denominator*** *(or the Least Common Multiple of the denominators), denoted by* $lcd(\boldsymbol{v})$*, is the smallest positive integer number divisible by all the denominators* q_j*, with* $j \in \{1, 2, \ldots, k\}$*.*

From rational to integer input data

- Multiplying a linear constraint by a positive number does not change the set of points satisfying it
- Multiplying the objective function by a positive number does not change the set of optimal solutions
- Accordingly, an LP problem with **rational input data** is equivalent to an LP problem with **integer input data**

The equivalent LP problem with integer input data

Given an LP problem in canonical maximization form with rational input data, the equivalent LP problem with integer input data reads as follows:

$$\frac{1}{lcd(\boldsymbol{c})} \max \quad \sum_{j=1}^{n} lcd(\boldsymbol{c})\, c_j\, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} lcd([\boldsymbol{a}_i \;\; b_i])\, a_{ij}\, x_j \leq lcd([\boldsymbol{a}_i \;\; b_i])\, b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \qquad \forall j \in \{1, 2, \ldots, n\},$$

where $\boldsymbol{a}_i$ denotes the i-th row of the constraint matrix $\boldsymbol{A}$.

Example: equivalent LP problem with integer input data (1/3)

Consider the following LP problem in canonical maximization form with $n = 5$ non-negative variables, $m = 1$ less-than-or-equal-to linear constraint and rational input data:

$$\begin{aligned} \max \ & \tfrac{1}{3} x_1 - \tfrac{11}{27} x_2 + \tfrac{7}{10} x_3 + \tfrac{7}{5} x_4 + \tfrac{2}{9} x_5 \\ \text{subject to } & \tfrac{1}{2} x_1 + \tfrac{3}{4} x_2 + \tfrac{11}{6} x_3 - \tfrac{1}{4} x_4 + \tfrac{5}{2} x_5 \leq \tfrac{25}{12} \\ & x_1, \quad x_2, \quad x_3, \quad x_4, \quad x_5 \geq 0 \end{aligned}$$

Example: equivalent LP problem with integer input data (2/3)

We have:

$$\underbrace{lcd\left(\begin{pmatrix}\frac{1}{3}\\ -\frac{11}{27}\\ \frac{7}{10}\\ \frac{7}{5}\\ \frac{2}{9}\end{pmatrix}\right)}_{=lcd(\boldsymbol{c})} = 270 \quad \text{and} \quad \underbrace{lcd\left(\begin{pmatrix}\frac{1}{2}\\ \frac{3}{4}\\ \frac{11}{6}\\ -\frac{1}{4}\\ \frac{5}{2}\\ \frac{25}{12}\end{pmatrix}\right)}_{=lcd\left(\begin{bmatrix}\boldsymbol{a}_1 & \frac{25}{12}\end{bmatrix}\right)} = 12$$

Example: equivalent LP problem with integer input data (3/3)

Accordingly, the equivalent LP problem with the same variables and constraints and with integer input data reads as follows:

$$
\begin{array}{lrrrrrrrrrrrrrr}
\frac{1}{270} & \max & 90 & x_1 & - & 110 & x_2 & + & 189 & x_3 & + & 378 & x_4 & + & 60 & x_5 \\
\text{subject to} & & 6 & x_1 & + & 9 & x_2 & + & 22 & x_3 & - & 3 & x_4 & + & 30 & x_5 & \leq & 25 \\
& & & x_1, & & & x_2, & & & x_3, & & & x_4, & & & x_5 & \geq & 0
\end{array}
$$

- The objective function is multiplied by $lcd(\boldsymbol{c}) = 270$ and the constraint by $lcd([\boldsymbol{a}_1 \; b_1]) = 12$

Summary (1/2)

Equivalent LP problems:

- from an optimal solution of one, an optimal solution of the other is directly obtained, and they have the same outcome

Objective function:

- $\min \boldsymbol{c}'\boldsymbol{x} = -\max(-\boldsymbol{c})'\boldsymbol{x}$: same optimal solutions, opposite optimal values

Constraints:

- "$\geq$" $\rightarrow$ "$\leq$" by multiplying by -1; "$=$" $\rightarrow$ two "$\leq$"
- "$\leq$" / "$\geq$" $\rightarrow$ "$=$" by adding a slack / subtracting a surplus variable

Summary (2/2)

Variables:

- $x_j \leq 0$: set $x_j = -x_j^+$; $x_j \gtreqless 0$: set $x_j = x_j^+ - x_j^-$, in the objective function and in every constraint

Two maximization forms:

- every generic LP problem can be rewritten in canonical and in standard maximization form
- in matrix notation, the standard form obtained with slack variables has constraint matrix $\begin{pmatrix} \boldsymbol{A} & \boldsymbol{I} \end{pmatrix}$

Rational input data:

- multiplying by the least common denominators gives an equivalent LP problem with integer data

Summary (1/2)

Reference forms:

- the **canonical** ($\max \boldsymbol{c}'\boldsymbol{x}$, $\boldsymbol{A}\,\boldsymbol{x} \leq \boldsymbol{b}$, $\boldsymbol{x} \geq 0$) and the **standard** ($\max \boldsymbol{c}'\boldsymbol{x}$, $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq 0$) maximization forms

Possible outcomes:

- every LP problem is either infeasible, or unbounded, or it admits an optimal solution with a finite optimal value
- an unbounded feasible region does not imply an unbounded LP problem
- the optimal solutions are either exactly one or infinitely many

Summary (2/2)

Equivalent formulations:

- minimization $\rightarrow$ maximization by changing the sign of $\boldsymbol{c}$
- "$\geq$" and "$=$" constraints $\rightarrow$ "$\leq$" constraints; "$\leq$" and "$\geq$" constraints $\rightarrow$ "$=$" constraints with slack and surplus variables
- non-positive and free variables $\rightarrow$ non-negative variables

Consequence:

- every generic LP problem can be rewritten in canonical and in standard maximization form, and with integer input data when its data are rational
- from now on, it is enough to study LP problems in these two forms